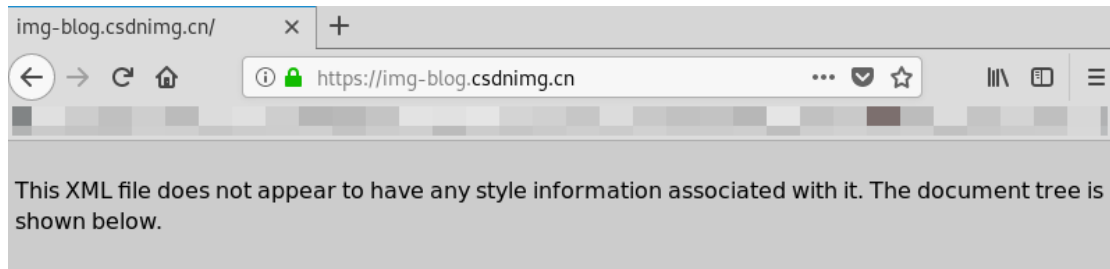


一、实验步骤

1. 访问 csdn 的图片站点 (<https://img-blog.csdnimg.cn/>), 并抓包, 保存为 https_csdn.pcapng



```
<Error>
  <Code>AccessDenied</Code>
  <Message>The bucket you access does not belong to you.</Message>
  <RequestId>5DC288A1B41F133534BE55F5</RequestId>
  <HostId>csdn-img-blog.oss-cn-beijing.aliyuncs.com</HostId>
</Error>
```

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	192.168.247.128	192.168.247.2	DNS	79	Standard query 0x7c09 A img-b1
2	0.000052708	192.168.247.128	192.168.247.2	DNS	79	Standard query 0x690f AAAA img
3	0.001765214	192.168.247.2	192.168.247.128	DNS	79	Standard query response 0x690f
4	0.002462414	192.168.247.2	192.168.247.128	DNS	550	Standard query response 0x7c09
5	0.002920983	192.168.247.128	36.249.73.239	TCP	74	41724 → 443 [SYN] Seq=0 Win=64
6	0.024843186	36.249.73.239	192.168.247.128	TCP	60	443 → 41724 [SYN, ACK] Seq=0 A
7	0.024880415	192.168.247.128	36.249.73.239	TCP	54	41724 → 443 [ACK] Seq=1 Ack=1
8	0.026676715	192.168.247.128	36.249.73.239	TLSv1...	571	Client Hello
9	0.027343937	36.249.73.239	192.168.247.128	TCP	60	443 → 41724 [ACK] Seq=1 Ack=51
10	0.053395030	36.249.73.239	192.168.247.128	TLSv1...	1514	Server Hello
11	0.053428291	192.168.247.128	36.249.73.239	TCP	54	41724 → 443 [ACK] Seq=518 Ack=
12	0.053473776	36.249.73.239	192.168.247.128	TCP	1514	443 → 41724 [PSH, ACK] Seq=146
13	0.053481852	192.168.247.128	36.249.73.239	TCP	54	41724 → 443 [ACK] Seq=518 Ack=
14	0.053500620	36.249.73.239	192.168.247.128	TLSv1...	394	Certificate, Server Key Exchan
15	0.053506173	192.168.247.128	36.249.73.239	TCP	54	41724 → 443 [ACK] Seq=518 Ack=
16	0.055222159	192.168.247.128	36.249.73.239	TLSv1...	147	Client Key Exchange, Change C1

2. 为了方便分析, 直接过滤 443 端口

No.	Time	Source	Destination	Protocol	Length	Info
5	0.002920983	192.168.247.128	36.249.73.239	TCP	74	41724 → 443 [SYN] Seq=0 Win=64
6	0.024843186	36.249.73.239	192.168.247.128	TCP	60	443 → 41724 [SYN, ACK] Seq=0 A
7	0.024880415	192.168.247.128	36.249.73.239	TCP	54	41724 → 443 [ACK] Seq=1 Ack=1
8	0.026676715	192.168.247.128	36.249.73.239	TLSv1...	571	Client Hello
9	0.027343937	36.249.73.239	192.168.247.128	TCP	60	443 → 41724 [ACK] Seq=1 Ack=51
10	0.053395030	36.249.73.239	192.168.247.128	TLSv1...	1514	Server Hello
11	0.053428291	192.168.247.128	36.249.73.239	TCP	54	41724 → 443 [ACK] Seq=518 Ack=
12	0.053473776	36.249.73.239	192.168.247.128	TCP	1514	443 → 41724 [PSH, ACK] Seq=146
13	0.053481852	192.168.247.128	36.249.73.239	TCP	54	41724 → 443 [ACK] Seq=518 Ack=
14	0.053500620	36.249.73.239	192.168.247.128	TLSv1...	394	Certificate, Server Key Exchan
15	0.053506173	192.168.247.128	36.249.73.239	TCP	54	41724 → 443 [ACK] Seq=518 Ack=
16	0.055222159	192.168.247.128	36.249.73.239	TLSv1...	147	Client Key Exchange, Change C1
17	0.055432024	36.249.73.239	192.168.247.128	TCP	60	443 → 41724 [ACK] Seq=3261 Ack
23	0.077580647	36.249.73.239	192.168.247.128	TLSv1...	365	New Session Ticket, Change C1

二、流量分析

1. 客户端向 443 端口发送 SYN 信号, 请求连接

```
5 0.002920983 192.168.247.128 36.249.73.239 TCP 74 41724 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1
```

2. 服务器回应连接

6 0.024843186 36.249.73.239 192.168.247.128 TCP 60 443 → 41724 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460

3. TCP/IP 三次握手完成

7 0.024880415 192.168.247.128 36.249.73.239 TCP 54 41724 → 443 [ACK] Seq=1 Ack=1 Win=64240 Len=0

4. Client Hello 信号，客户端发送随机数字+自己可以支持的加密方法

8 0.026676715 192.168.247.128 36.249.73.239 TLSv1... 571 Client Hello
9 0.027343937 36.249.73.239 192.168.247.128 TCP 60 443 → 41724 [ACK] S

Transport Layer Security

TLSv1.2 Record Layer: Handshake Protocol: Client Hello

Content Type: Handshake (22)

Version: TLS 1.0 (0x0301)

Length: 512

Handshake Protocol: Client Hello

Handshake Type: Client Hello (1)

Length: 508

Version: TLS 1.2 (0x0303) 支持的协议版本

Random: 099dbc5aac040842114bfc51ec9e36c9816e62264fdc65... 客户端生成的随机数1
用于稍后生成对话密钥

Session ID Length: 32

Session ID: 6300bc16c2c6ce10216c68247561ff35fe7fed4f75c4a005...

Cipher Suites Length: 28

Cipher Suites (14 suites)

Cipher Suite: TLS_AES_128_GCM_SHA256 (0x1301)

Cipher Suite: TLS_CHACHA20_POLY1305_SHA256 (0x1303) 支持的加密算法

Cipher Suite: TLS_AES_256_GCM_SHA384 (0x1302)

Cipher Suite: TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256 (0xc02b)

Cipher Suite: TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256 (0xc02f)

Cipher Suite: TLS_ECDHE_ECDSA_WITH_CHACHA20_POLY1305_SHA256 (0xcca9)

Cipher Suite: TLS_ECDHE_RSA_WITH_CHACHA20_POLY1305_SHA256 (0xcca8)

Cipher Suite: TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384 (0xc02c)

Cipher Suite: TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384 (0xc030)

Cipher Suite: TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256 (0xc02e)

5. 服务器 ACK 应答

9 0.027343937 36.249.73.239 192.168.247.128 TCP 60 443 → 41724 [ACK] Seq=1 Ack=518 Win=64240 Len=0

6. Server Hello 信号，服务器发送随机数字+选择双方都支持的加密方式，这里选择的是 TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256

10 0.053395030 36.249.73.239 192.168.247.128 TLSv1... 1514 Server Hello
11 0.053428291 192.168.247.128 36.249.73.239 TCP 54 41724 → 443 [ACK]
12 0.053473776 36.249.73.239 192.168.247.128 TCP 1514 443 → 41724 [PSH,
13 0.053481852 192.168.247.128 36.249.73.239 TCP 54 41724 → 443 [ACK]

Transport Layer Security

TLSv1.2 Record Layer: Handshake Protocol: Server Hello

Content Type: Handshake (22)

Version: TLS 1.2 (0x0303)

Length: 74

Handshake Protocol: Server Hello

Handshake Type: Server Hello (2)

Length: 70

Version: TLS 1.2 (0x0303) 确认协议版本

Random: 80b40e1ba7600a5cd9a4ab3845edb34e2e344cb1e581044f... 服务器生成的随机数2
用于稍后生成对话密钥

Session ID Length: 0

Cipher Suite: TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256 (0xc02f) 确认加密算法

Compression Method: null (0)

Extensions Length: 30

7. Certificate: 服务器发送自己的公钥和从 CA 申请的证书;

Server Key Exchange: 告诉客户端开始转换成对称密钥;

Server Hello Done: Server Hello 结束

14	0.053500620	36.249.73.239	192.168.247.128	TLSv1...	394	Certificate, Server Key Exchange, Server Hello Done
15	0.0535006173	192.168.247.128	36.249.73.239	TCP	54 41724 → 443	[ACK] Seq=518 Ack=3261 Win=62440 Len=0
- Frame 14: 394 bytes on wire (3152 bits), 394 bytes captured (3152 bits) on interface 0 - Ethernet II, Src: Vmware_f0:80:5f (00:50:56:f0:80:5f), Dst: Vmware_c4:d3:f5 (00:0c:29:c4:d3:f5) - Internet Protocol Version 4, Src: 36.249.73.239, Dst: 192.168.247.128 - Transmission Control Protocol, Src Port: 443, Dst Port: 41724, Seq: 2921, Ack: 518, Len: 340 [3 Reassembled TCP Segments (2867 bytes): #10(1381), #12(1460), #14(26)] - Transport Layer Security - TLSv1.2 Record Layer: Handshake Protocol: Certificate - Content Type: Handshake (22) - Version: TLS 1.2 (0x0303) - Length: 2862 - Handshake Protocol: Certificate - Handshake Type: Certificate (11) - Length: 2858 - Certificates Length: 2855 - Certificates (2855 bytes) - Certificate Length: 1682 - Certificate: 3082068e30820576a003020102021007cc642414bac5e3f0... (id-at-commonName=*.csdnimg.cn,id-at-organization - Certificate Length: 1167 - Certificate: 3082048b30820373a00302010202100546fe1823f7e1941d... (id-at-commonName=GeoTrust RSA CA 2018,id-at-orga						

8. **client_key_exchange**: 交换 DH 算法的参数, 此时客户端和服务端的对称算法密钥达成一致。

change_cipher_spec: 客户端发送 Change cipher 信号

Encrypted Handshake Message: 发送用对称加密的第一个数据, 也就是 finish 信号

16	0.055222159	192.168.247.128	36.249.73.239	TLSv1...	147	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
17	0.055432024	36.249.73.239	192.168.247.128	TCP	60 443 → 41724	[ACK] Seq=3261 Ack=611 Win=64240 Len=0
23	0.077580647	36.249.73.239	192.168.247.128	TLSv1...	365	New Session Ticket, Change Cipher Spec, Encrypted Handshake Message,
24	0.077604066	192.168.247.128	36.249.73.239	TCP	54 41724 → 443	[ACK] Seq=611 Ack=3572 Win=62780 Len=0
- Transport Layer Security - TLSv1.2 Record Layer: Handshake Protocol: Client Key Exchange - Content Type: Handshake (22) - Version: TLS 1.2 (0x0303) - Length: 37 - Handshake Protocol: Client Key Exchange - Handshake Type: Client Key Exchange (16) - Length: 33 - EC Diffie-Hellman Client Params - TLSv1.2 Record Layer: Change Cipher Spec Protocol: Change Cipher Spec - Content Type: Change Cipher Spec (20) - Version: TLS 1.2 (0x0303) - Length: 1 - Change Cipher Spec Message - TLSv1.2 Record Layer: Handshake Protocol: Encrypted Handshake Message - Content Type: Handshake (22) - Version: TLS 1.2 (0x0303) - Length: 40 - Handshake Protocol: Encrypted Handshake Message						

9. **New session ticket**: 目的是为了之后不再进行握手, 节约服务器资源;

change_cipher_spec: 服务器发送 Change cipher 信号

Encrypted Handshake Message: 发送用对称加密的第一个数据

Application Data: 数据加密后的密文

23	0.077580647	36.249.73.239	192.168.247.128	TLSv1...	365	New Session Ticket, Change Cipher Spec, Encrypted Handshake Message, Application Data
24	0.077604066	192.168.247.128	36.249.73.239	TCP	54 41724 → 443	[ACK] Seq=611 Ack=3572 Win=62780 Len=0
31	0.197194690	192.168.247.128	36.249.73.239	TLSv1...	231	Application Data
- Transport Layer Security - TLSv1.2 Record Layer: Handshake Protocol: New Session Ticket - Content Type: Handshake (22) - Version: TLS 1.2 (0x0303) - Length: 186 - Handshake Protocol: New Session Ticket - Handshake Type: New Session Ticket (4) - Length: 182 - TLS Session Ticket - TLSv1.2 Record Layer: Change Cipher Spec Protocol: Change Cipher Spec - Content Type: Change Cipher Spec (20) - Version: TLS 1.2 (0x0303) - Length: 1 - Change Cipher Spec Message - TLSv1.2 Record Layer: Handshake Protocol: Encrypted Handshake Message - Content Type: Handshake (22) - Version: TLS 1.2 (0x0303) - Length: 40 - Handshake Protocol: Encrypted Handshake Message - TLSv1.2 Record Layer: Application Data Protocol: http2 - Content Type: Application Data (23) - Version: TLS 1.2 (0x0303) - Length: 64 - Encrypted Application Data: db6a20d0738edc18823e97b200c4507a1d0875231d4f235d...						

10. 服务器发送 Encrypted Alert 信号, 结束通信

146	0.247026097	192.168.247.128	123.157.149.32	TLSv1...	77	Encrypted Alert
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11. TCP/IP 四次挥手, 断开连接

54	7.440110357	192.168.247.128	36.249.73.239	TCP	54 41724 → 443	[FIN, ACK] Seq=1207 Ack=4914 Win=62780 Len=0
55	7.440224723	36.249.73.239	192.168.247.128	TCP	60 443 → 41724	[ACK] Seq=4914 Ack=1208 Win=64239 Len=0
58	7.462360881	36.249.73.239	192.168.247.128	TCP	60 443 → 41724	[FIN, PSH, ACK] Seq=4914 Ack=1208 Win=64239 Len=0
59	7.462399246	192.168.247.128	36.249.73.239	TCP	54 41724 → 443	[ACK] Seq=1208 Ack=4915 Win=62780 Len=0